Masterplan: End-to-End Private LoRaWAN Smart Agriculture System

Executive Summary

This document presents the complete hardware, system architecture, and deployment masterplan for establishing a **private, enterprise-grade LoRaWAN Internet of Things (IoT) network** for agricultural monitoring.

The system operates on a **two-tier architecture**:

1. **The Gateway Infrastructure ("The Tower"):** A central base station that captures long-range wireless telemetry from across the property and routes it to cloud/local servers.
2. **The WisBlock Field Sensor Nodes ("The Field Workers"):** Weatherproof, solar-powered multi-sensor stations deployed directly in crop fields, orchards, and greenhouses to collect real-time crop, soil, and microclimate data.

By combining an in-house Gateway with WisBlock LoRaWAN sensor nodes, the farm achieves multi-kilometer wireless coverage without monthly cellular subscription fees per sensor.

Complete System Architecture Diagram

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┌──────────────────────────────────────────────────────────────────────────────────────┐
│                                    FIELD SENSOR LEVEL                                │
│                                                                                      │
│  [WisBlock Node #1]        [WisBlock Node #2]         [WisBlock Node #3]       [WisBlock Node #N]│
│   (Soil & Microclimate)    (Rain & UV Index)         (Weather Station)       (Modbus/SDI-12)  │
│           │                         │                          │                         │    │
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            │                         │                          │                         │
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                                                   │
                                                   │  LoRaWAN Radio Signals
                                                   │  (AS923 Frequency Band / 900-928 MHz)
                                                   │  Range: Up to 5-15 km Line-of-Sight
                                                   ▼
┌──────────────────────────────────────────────────────────────────────────────────────┐
│                              GATEWAY / INFRASTRUCTURE LEVEL                          │
│                                                                                      │
│  ┌───────────────────────────────────────────────────────────────────────────────┐  │
│  │ Outdoor High-Gain Antenna (900-930 MHz) + GPS Antenna                         │  │
│  └───────────────────────────────────────────┬───────────────────────────────────┘  │
│                                              │ RF Pigtails                          │
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│  │ RAK5146 LoRaWAN Concentrator Card (SPI Interface)                             │   │
│  └───────────────────────────────────────────┬───────────────────────────────────┘  │
│                                              │ WisLink Pi HAT Adapter Board         │  │
│  ┌───────────────────────────────────────────┴───────────────────────────────────┐  │
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│ │ Raspberry Pi 4 Single-Board Computer (Gateway Brain)                    │ │
│ └─────────────────────────────────────────────────────────────────────────┘ │
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                                  │
                                  │  Ethernet / IP Backhaul (Cat5e / Cat6 Cable)
                                  ▼
┌─────────────────────────────────────────────────────────────────────────────┐
│                        APPLICATION & SERVER LEVEL                           │
│                                                                             │
│ ┌─────────────────────────────┐        ┌──────────────────────────────────┐ │
│ │ ChirpStack / TTN Network Server │ ──────> │ Visualization Dashboard              │ │
│ │ (Decodes encrypted payloads)    │         │ (Graphs, Alerts, Automated Irrigation Rules) │ │
│ └─────────────────────────────┘        └──────────────────────────────────┘ │
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Part 1: Gateway Master Checklist (Network Infrastructure)

The Gateway acts as the central hub. It listens on 8 radio channels simultaneously to pick up signals from thousands of field sensors and forwards that data to the server via Ethernet.

Component	Description	Primary Function	Crucial Purchasing / Technical Notes
Raspberry Pi 4 Model B	Main host computer (2GB, 4GB, or 8GB RAM).	Runs gateway bridge software, handles network packet routing.	Requires official 5V/3A USB-C power supply to avoid power drops during radio spikes.
RAK5146 Concentrator Module	High-performance mPCIe LoRaWAN gateway card (Semtech SX1303).	Receives radio packets from sensors on multiple channels simultaneously.	**MUST BE: SPI Version & 900–928 MHz (AS923/US915).** Do *not* buy 868 MHz or USB variants.
RAKwireless WisLink Pi HAT	Interface adapter board.	Physical bridge connecting the RAK5146 mPCIe card to the Raspberry Pi GPIO pins.	Includes slot for mPCIe card and hardware mounting points.
Sub-GHz Outdoor Antenna	Fiber-glass omnidirectional antenna (900–930 MHz tuned).	Broadcasts and receives LoRa radio waves across farm terrain.	**CRITICAL:** NEVER power on the Pi without the antenna connected, or the radio chip will burn out.
Active GPS Antenna	GPS positioning antenna (u.FL/SMA connection).	Provides accurate time synchronization (Time Difference of Arrival) for network packets.	Plugs into the GPS port on the RAK5146 concentrator card.
2x RF Pigtail Cables	u.FL / IPEX to SMA Female pigtail jumpers.	Adapts tiny radio connectors on the RAK5146 board to standard external SMA antenna ports.	1x for LoRa antenna, 1x for GPS antenna.
Official Pi Power Supply	5V / 3A USB-C Power Adapter.	Supplies continuous, clean electrical power to the Pi and radio HAT.	Generic phone chargers cause system crashes during packet transmission.

MicroSD Card & Reader	32GB or 64GB Class 10 / High Endurance MicroSD card + USB flasher.	Holds the Gateway Linux OS and LoRa Packet Forwarder software.	High-endurance cards prevent data corruption from continuous log writing.
Ethernet Cable	Cat5e or Cat6 heavy-duty network patch cable.	Connects gateway to local router / internet backhaul.	Provides stable connection back to the network server.
Mounting Hardware & Enclosure	Brass standoffs, M2.5 & M2 screws, IP67 waterproof outdoor box.	Structurally locks components together and seals hardware against weather.	Needed for mounting on elevated tower, roof, or field pole.

Part 2: WisBlock Agriculture Kit Breakdown (Field Sensor Nodes)

The WisBlock kit provides modular hardware to construct modular field stations. Each node consists of a Core Processor, a Base Board, and targeted selection of sensors.

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│             RAK4631 Core Module (MCU + LoRa)            │
└────────────────────────────┬────────────────────────────┘
                             │ Plugs into
┌────────────────────────────▼────────────────────────────┐
│      RAK19007 / RAK19001 WisBlock Carrier Board         │
└─┬──────────────┬──────────────┬──────────────┬──────────┘
  │ Slot A       │ Slot B       │ Slot C       │ Slot D
  ▼              ▼              ▼              ▼
[Soil Probe]   [Temp/Hum]     [Rain Sensor]  [RS485/Modbus]
```

1. Base Boards & Core Processors (Brains & Connectivity)

- **RAK4631 — WisBlock Core Module (MCU + LoRaWAN + BLE)**
  - **Function:** Main microcontroller (Nordic nRF52840) paired with a Semtech SX1262 LoRa transceiver.
  - **Agricultural Application:** Executes code, gathers sensor measurements, encrypts the data, and broadcasts it over LoRaWAN (AS923) to the Gateway.
- **RAK19007 — WisBlock Base Board (Standard)**
  - **Function:** Compact carrier board offering 4 sensor slots, 1 IO slot, USB-C programming port, and integrated solar panel / lithium battery charging circuit.
  - **Agricultural Application:** Serves as the foundation for standard 4-sensor field nodes (e.g., soil + microclimate monitoring station).
- **RAK19001 — WisBlock Dual IO Base Board (Expanded)**
  - **Function:** Large-format carrier board providing 6 sensor slots and 2 IO slots.
  - **Agricultural Application:** Designed for master weather nodes that require high sensor density (e.g., full weather station + multiple depth probes).

2. Environmental & Soil Sensors (Data Collection)

- **RAK12023 / RAK12035 — Soil Moisture & Temperature Sensor**
  - **Function:** Capacitive moisture probe and controller interface.
  - **Agricultural Application:** Measures volumetric soil water content and ground temperature without metal corrosion, optimizing irrigation scheduling.
- **RAK1906 — Integrated Environmental Sensor (BME680)**
  - **Function:** 4-in-1 sensor measuring air temperature, relative humidity, barometric pressure, and air quality (VOCs).
  - **Agricultural Application:** Tracks microclimates across canopy levels, calculates dew point to predict frost, and monitors ambient conditions.
- **RAK12019 — UV Light Sensor (LTR-390UV)**
  - **Function:** Measures ambient UV Index (UVI) and ultraviolet radiation intensity.
  - **Agricultural Application:** Monitors solar stress on fruit skin and dictates deployment of shade cloth systems.
- **RAK12010 / RAK1903 — Ambient Light Sensors (VEML7700 / OPT3001)**
  - **Function:** Precision Lux meters measuring sunlight levels across wide dynamic ranges.
  - **Agricultural Application:** Evaluates Photosynthetically Active Radiation (PAR) potential for photosynthesis inside greenhouses or under shade nets.
- **RAK12011 — Waterproof Barometric Pressure Sensor (LPS33HW)**
  - **Function:** Fully sealed, water-impermeable air pressure sensor.
  - **Agricultural Application:** Predicts incoming weather events (sudden pressure drops) and measures liquid level/pressure in water tanks.
- **RAK12005 / RAK12030 — Rain Detection Sensor Module**
  - **Function:** Surface-precipitation grid board.
  - **Agricultural Application:** Immediately detects onset of rain to pause automatic irrigation cycles or trigger greenhouse vent closure.

3. Industrial Protocol Interfaces (External Expansion)

- **RAK5802 — RS485 / Modbus Communication Module**
  - **Function:** Converts RS485 serial communications to microcontroller readable signals.
  - **Agricultural Application:** Connects field nodes to heavy-duty industrial multi-depth soil probes, weather instruments, or water flow meters.
- **RAK5801 — 4–20mA Analog Current Loop Module**
  - **Function:** Translates standard industrial 4–20mA current signals to digital values.
  - **Agricultural Application:** Interfaces with legacy industrial sensors, deep-well pressure transducers, and chemical dosing systems.
- **RAK13010 — SDI-12 Communication Module**
  - **Function:** Specialized SDI-12 protocol converter.
  - **Agricultural Application:** Connects to scientific grade agronomy probes (e.g., multi-segment soil moisture profile spears, sap flow sensors).

4. Optional Cellular Modules (Direct-to-Cloud Options)

> **Note:** These optional expansion cards plug into WisBlock base boards *only* if bypassing the LoRaWAN gateway.

- **RAK5860 — NB-IoT / LTE-M Cellular Module (Quectel BG77):** Connects node directly to cellular towers via 4G/5G SIM card. Recommended *only* for standalone nodes deployed far beyond LoRaWAN radio range.
- **RAK13101 — GSM / GPRS Module (Quectel MC20CE):** Legacy 2G cellular module. **Not recommended** due to ongoing global 2G network shutdowns.

Part 3: Interoperability & Integration Matrix

Operating Parameter	LoRaWAN Gateway (List #1)	WisBlock Agriculture Nodes (List #2)	Compatibility Rule
System Role	Base Station / Infrastructure	End-Device / Telemetry Field Node	Gateway collects and forwards data from multiple nodes.
Frequency Band	AS923 / 900–928 MHz	AS923 / 900–928 MHz	**MUST MATCH EXACTLY.** Both items must be ordered in the 900MHz range.
Communication Link	Packet Forwarder over Ethernet/IP	LoRa Radio Payloads	Sensor nodes encrypt data; Gateway passes encrypted packets to server.
Power Source	Main AC Power + 5V/3A Supply	3.7V Li-Ion Battery + Solar Panel	Gateway runs 24/7; Field nodes run autonomously via solar harvesting.
Cellular Option	Not required (uses Ethernet)	**Select "None"**	Avoids purchasing unnecessary cellular chips or paying monthly SIM fees.

Part 4: Procurement & Ordering Quick-Guide

When placing orders, verify the following configuration flags to prevent ordering mismatched hardware:

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                  │       ORDERING CHECKLIST       │
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                                  │
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           ▼                                            ▼
┌────────────────────────────┐  ┌────────────────────────────┐
│   GATEWAY HARDWARE CHECK   │  │ WISBLOCK KIT SELECTION CHECK│
├────────────────────────────┤  ├────────────────────────────┤
│ [✓] RAK5146: SPI Interface │  │ [✓] Frequency: 900 MHz (AS923)│
│ [✓] Frequency: 900-928 MHz (AS923) │ │ [✓] Cellular Option: "None" │
│ [✓] Accessories: 2x Pigtails + GPS │ │     (Saves ~$54/kit over cellular)│
│ [✓] Power: Official 5V/3A Adapter  │ │                            │
└────────────────────────────┘  └────────────────────────────┘
```

Procurement "Red Flag" Summary

1. **Frequency Matching:** Ensure all LoRa radios (RAK5146 concentrator and RAK4631 core modules) are ordered for **900–928 MHz / AS923**. European frequencies (868 MHz) will not communicate and violate radio regulations in AS923 regions.
2. **Cellular Variant on Kit:** When ordering the WisBlock Agriculture Kit, select **"None"** under cellular modules. Since the gateway handles backhaul, adding cellular modules adds unnecessary cost and power consumption.
3. **Gateway Interface:** Ensure the RAK5146 card is the **SPI version** to match the WisLink Pi HAT GPIO configuration.

Part 5: Deployment & Commissioning Steps

1. **Assemble Gateway Stack:**
   - Secure the WisLink Pi HAT onto the Raspberry Pi 4 GPIO header using brass standoffs.
   - Seat the RAK5146 card into the mPCIe slot and secure with M2 screws.
   - Attach both RF pigtails and connect the external Sub-GHz and GPS antennas before powering on.
2. **Configure Network Server:**
   - Flash Raspberry Pi OS onto the High Endurance MicroSD card.
   - Install packet forwarder software (e.g., ChirpStack Gateway OS or Semtech UDP Packet Forwarder).
   - Connect Ethernet cable and verify connection to local server or cloud endpoint.
3. **Assemble WisBlock Nodes:**
   - Plug RAK4631 Core into slot C of the RAK19007/RAK19001 base board.
   - Connect required environmental and soil sensors into slots A, B, D.

- Connect rechargeable lithium battery and solar panel to the designated base board JST headers.

4. **Provision and Field Test:**
    - Register node DevEUI, AppEUI, and AppKey on your LoRaWAN Network Server.
    - Power on node and verify OTAA (Over-The-Air Activation) join request on the Gateway logs.
    - Enclose node in an IP67 weatherproof enclosure and mount in crop target area.